Intrinsic Self-Correction Requires Extrinsic Direction

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Abstract

Large language models are increasingly asked to revise their own outputs over multiple conversation turns, often by users who cannot specify what should change. Prior work on self-correction has largely examined single rounds of self-critique on tasks with verifiable answers, finding that models improve mainly when external feedback is available. Whether repeated, undirected revision requests help or harm output quality remains unexamined. We study this setting directly, running five-turn revision conversations with six LLMs on 40 tasks across five domains, from code and data reasoning to business and creative writing. Under undirected requests, genuine revision declines from 39% of responses at the second turn to 13% by the fifth, with models instead restating or declining in ways that leave outputs effectively unchanged. Revisions that do occur reduce quality by 0.74 levels on a five-point scale, and every model's first draft rates highest. A single targeted critique reverses this pattern, improving quality by 1.16 levels. These results suggest that intrinsic self-correction fails not from lack of capacity but from lack of direction: models revise effectively only when told what is wrong.

1 Introduction

Large language models are evaluated, for the most part, alone. A model is handed a fully specified task, produces a single answer, and is scored against a reference, and on benchmarks of this form current models perform well (Liang et al., 2023). The tasks that motivate their deployment, however, are rarely solved in a single exchange. Increasingly they are workplace tasks carried out in collaboration with a user who supplies the goal, reacts to the output, and asks for changes (Anthropic, 2025a; Laban et al., 2025). In this setting the quality a user ultimately receives is not a property of the model alone but of the interaction, and it depends heavily on what the user is able to contribute to it. What users contribute is often thin: real prompts are frequently underspecified, and underspecification measurably degrades the outputs that follow (Yang et al., 2025). Studying this dependence of output quality on user-side behavior is the concern of human-AI interaction, and it is where we situate this work.

One user-side contribution matters more than most: the ability to say what is wrong with an output. A user who can identify a specific weakness can direct the model to fix it. A user who cannot, either because the task exceeds their expertise or because the output looks acceptable but they are unsure, has a single move available. This is the ordinary condition of a non-expert working above the level at which they could specify a correction, a difficulty that scalable-oversight research identifies as central: users often cannot articulate precise intent or reliably validate a complex output (Zhou et al., 2026). They ask the model to improve the work without saying what improving it would mean. This undirected request, some version of *make it better*, is among the most common things people ask of these systems, and vendor guidance actively encourages it, advising users to treat the model as a collaborator and to iterate toward a better result (OpenAI, 2025a; Anthropic, 2025c). It is also, we find, where the collaboration quietly fails.

Whether models can improve their own outputs has been studied under the heading of self-refinement. Madaan et al. (2024) show that a model given the chance to critique and rewrite its work can raise its quality, and locate the benefit in feedback that is *actionable* and *specific*: that names a concrete change to a concrete part of the output. Subsequent work finds the benefit fragile. Huang et al. (2024) report that without external feedback models struggle to correct their own reasoning and at times degrade it, and that apparent gains often trace to information leaked through oracle labels or more informative prompts. A critical survey by Kamoi et al. (2024) reaches a compatible conclusion across tasks, locating the bottleneck in feedback generation rather than in the capacity to revise. In parallel, Laban et al. (2025) show that models lose accuracy across multi-turn conversations when task requirements are revealed gradually, a degradation they attribute to underspecification accumulating over turns.

These results bracket the question we ask but do not answer it. Revision driven only by the model's own judgment, with no feedback supplied from outside, is what Kamoi et al. (2024) and Huang et al. (2024) term *intrinsic* self-correction, and the self-refinement literature studies it alongside revision that does carry a signal, whether generated by the model, supplied by an oracle, or embedded in the prompt, asking in each case whether that signal is enough to improve an output. The multi-turn literature studies accuracy on tasks whose requirements arrive piecewise. Neither measures what happens in the case a non-expert user actually creates: a fully specified task whose output is already adequate, followed by repeated requests to revise that carry no feedback at all. Measuring it, moreover, turns out to require controlling for an artifact that the revision act itself introduces into automated evaluation, which prior work has had no reason to address.

We study this case directly. A model completes one of forty fully specified tasks and is then offered, at each of four subsequent turns, a neutral choice to keep its output or revise it, with no indication of what to change. The design spans six models and yields 720 conversations and 3,600 model responses. We ask three questions. First, when a model is asked to revise without direction, does it revise, or decline while appearing to engage (RQ1)? Second, when it does revise, does the output improve or degrade (RQ2)? Third, if undirected revision fails, is the cause the model's capacity to revise or the absence of direction (RQ3)? To separate the effect from the way it is measured, we identify and strip the meta-commentary that models attach to revisions, and validate every quality judgment against human raters.

The answers are consistent across the six models. Asked to revise without direction, models mostly do not: genuine revision falls from 39% of responses at the second turn to 13% at the fifth, with the remainder being restatements and declines that change no task content. When a model does revise, the output tends to get worse, and for every model the quality-maximizing stopping point is the first turn. The failure is one of direction rather than capacity, in that a single specific critique reverses the decline and raises quality well above what undirected iteration reaches. We also find that the meta-commentary models wrap around revisions biases an automated judge in two opposing directions at once, inflating its preference for the earlier draft while lowering the measured quality of the later one, an effect large enough that leaving it uncontrolled changes the conclusion. Together these results identify a property that single-turn evaluation does not measure: whether a model, handed its own adequate output, will decline to make it worse. We argue this property deserves to be evaluated alongside first-turn capability, and that the practical remedy is to supply the direction the model cannot generate for itself: to withhold the request to revise until there is a specific weakness to direct it at. Intrinsic self-correction, on its own, is not enough; it requires extrinsic direction.

2 Related Work

Human-AI interaction and user-side determinants of quality. A growing body of work treats the quality of a model's output not as a fixed property of the model but as an outcome of the interaction in which it is produced, shaped by what the user contributes. Deployment data show that models are increasingly applied to open-ended workplace tasks rather than the closed, fully specified problems on which they are benchmarked (Anthropic, 2025a). In that setting the user's instructions are often thin: real prompts are frequently underspecified, and underspecification measurably degrades and destabilizes the outputs that follow (Yang et al., 2025). The difficulty is most acute for non-expert users, whom scalable-oversight research identifies as frequently unable to articulate precise intent or to validate a complex output (Zhou et al., 2026). Laban et al. (2025) show that this dependence compounds across turns, with accuracy falling when requirements are revealed gradually. We situate our work in this line and treat the revision request itself as a user-side variable, asking how its most impoverished form, a request to revise that carries no direction at all, affects output quality.

Over-reliance and automation bias. Whether such degradation is noticed depends on the user's ability to evaluate the output, and a long line of human-factors research indicates that ability is limited. Automation bias, the tendency to accept a machine's recommendation and to underscrutinize it, is documented across decades of studies and affects experts and novices alike (Skitka et al., 1999; Parasuraman and Manzey, 2010). In AI-assisted decision-making the same pattern holds: fluent explanations and confident presentation raise users' acceptance of model output without improving their ability to separate correct outputs from incorrect ones (Bansal et al., 2021; Buçinca et al., 2021). Reliance behaves as a cost-benefit trade-off in which users accept the model's answer rather than pay the cognitive cost of verifying it, so a degraded output can pass uncaught (Vasconcelos et al., 2023; Schemmer et al., 2023). This literature explains why the failure we document is easy to miss: a user who cannot distinguish a worse revision from a better one absorbs the loss, and the tokens spent reaching it are billed all the same.

Intrinsic self-correction and self-refinement. Whether models can improve their own outputs has been studied under the heading of self-refinement. Madaan et al. (2024) show that a model prompted to critique and rewrite its work can raise its quality, and locate the benefit in feedback that is *actionable* and *specific*, naming a concrete change to a concrete part of the output. Later work finds the benefit contingent on an external signal. Huang et al. (2024) report that without external feedback models struggle to correct their own reasoning and at times degrade it, and that apparent gains often trace to oracle labels or to more informative prompts. The limit appears to lie in error detection rather than error correction: Tyen et al. (2024) find that models cannot reliably locate their own reasoning errors but can fix them once the location is supplied, and Tsui (2025) document a self-correction blind spot in which models repair errors introduced by others while overlooking their own. Reliable self-correction correspondingly depends on a strong external verifier or on additional training rather than on intrinsic prompting (Zhang et al., 2024; Qu et al., 2024). A critical survey by Kamoi et al. (2024) synthesizes these results and locates the bottleneck in feedback generation rather than in the capacity to revise, the seam on which our finding sits (Kim et al., 2024; Shinn et al., 2023). This literature asks whether a given feedback signal is sufficient to improve an output; we ask the complementary question of what happens when the signal is absent entirely and the output to be revised is already adequate.

Termination and overthinking. A parallel failure appears within a single response. Reasoning models continue generating past the point at which their answer is complete and degrade it through redundant self-correction (Chen et al., 2024b, 2025), and test-time scaling produces non-monotonic accuracy that rises and then falls as further computation introduces variance (Ghosal et al., 2025). Knowing when to stop is now treated as an open problem in its own right (Sui et al., 2025). Inside a response this failure manifests as overthinking; across responses, we argue, it manifests as the revision cliff we measure. In both cases an optimal stopping point exists and the model overshoots it.

Sycophancy and compliance under pressure. The revision probe also exerts a mild but recurring pressure to comply, and models are known to yield to such pressure. Instruction-tuned models exhibit sycophancy, agreeing with users and avoiding disagreement even when holding firm would be more helpful (Perez et al., 2023; Sharma et al., 2024; Wei et al., 2024), a tendency embedded deeply enough in preference training that a 2025 model update was rolled back for it (OpenAI, 2025b). Under sustained multi-turn pressure the effect strengthens, with correct answers revised toward incorrect ones when a user pushes back or persists (Xu et al., 2024; Fanous et al., 2025). Our repeated revision offer is a weak instance of this pressure, and the cumulative degradation we observe is consistent with the same compliance mechanism operating across turns.

Biases in LLM-as-judge evaluation. Using LLMs to evaluate generated text is now standard practice (Zheng et al., 2024; Li et al., 2024), but LLM judges carry biases that our setting activates directly. Beyond the documented self-preference and position biases (Panickssery et al., 2024; Koo et al., 2024), judges reward stylistic and formatting cues over substance: verbose outputs score higher regardless of quality (Singhal et al., 2023), and surface features such as lists, boldface, and confident framing raise a judge's score independently of content (Wu and Aji, 2025; Zhang et al., 2025; Chen et al., 2024a; Ye et al., 2024). The meta-commentary that models attach to revisions is precisely such a cue, and we find it biases an automated judge strongly enough to reverse the measured direction of the effect. Any study that compares first-draft and revised outputs with an LLM judge inherits this confound, and we treat controlling for it as a methodological requirement rather than a robustness check.

Token efficiency and the cost of AI. The waste we quantify has an economic dimension. Enterprise inference spending has risen steeply, roughly doubling in under a year, and the multi-turn agentic workflows driving that growth are where undirected revision loops live (Menlo Ventures, 2025; Gartner, 2026; Deloitte AI Institute, 2026; McKinsey & Company, 2025). Our revision tax is adjacent to the single-turn over-generation metric of Borisov et al. (2026) but captures a distinct multi-turn cost, the tokens spent revising an already-sufficient output past the point of peak quality.

Multi-turn degradation and closest work. Quality loss across multiple turns has been observed before, though through different mechanisms. Early dialogue systems lose coherence over long conversations (Zhang et al., 2020; Thoppilan et al., 2022), a degradation driven by topic drift rather than by revision. Closest to our work, Laban et al. (2025) show that models lose accuracy across multi-turn conversations when a task's requirements are revealed piecewise, and attribute the loss to underspecification accumulating over turns as the model commits to early assumptions it cannot recover from. Our setting differs in a way that isolates a distinct cause: the task is fully specified from the first turn and the initial output is already sufficient, so the degradation we measure is produced by the revision act itself rather than by information arriving late. The two findings are complementary, one showing failure when requirements accumulate and the other showing failure when a sufficient output is revised without direction.

3 Methods

3.1 Design

We use a five-turn conversational design in which a model produces an initial output (Turn 1) and is then asked at each subsequent turn whether it would like to revise. The working probe is deliberately balanced: "Would you like to keep this as your final version, or would you like to revise it?" This offers an explicit off-ramp at every turn, unlike revision-implying probes that trigger near-universal compliance (see §3.8).

Domain	n	Example tasks
Code	8	Debounce function, email validator
Data logic	8	Discount stacking, scheduling
Analysis	8	Quarterly sales summary, policy memo
Writing	8	PTO request, LinkedIn announcement
Creative	8	Story opening, tone rewrite

Table 1: Task domains. Each domain contains 8 scenarios spanning typical enterprise and consumer AI use cases.

The experiment crosses 6 models $\times$ 40 scenarios $\times$ 3 runs = **720 trials** (3,600 model-turn observations). All generation uses temperature 1.0, to maximize output diversity and avoid ceiling effects from low-temperature determinism, consistent with standard practice for open-ended generation. Our design cannot fully separate the effect of the revision act from the prompt that offers it; a no-probe control would be needed to isolate the revision mechanism alone.

Models. Claude Sonnet 4 (Anthropic), GPT-4o (OpenAI), Gemini 2.5 Flash (Google), Llama 3.3 70B (Meta, via Together AI), Qwen 3 235B (Alibaba, via OpenRouter), and DeepSeek V4 (DeepSeek).

Scenarios. Forty scenarios span five task domains (Table 1): code (8 tasks), data logic (8), analysis (8), writing (8), and creative writing (8). Scenarios range from function implementation to policy memos to story openings.

3.2 Response Classification

At each turn $\geq$ 2, the model's response is classified as either a *genuine revision* (new task content produced) or a *meta-response* (the model declines, restates, or comments without substantive modification). An initial keyword-based classifier (checking for decline phrases like "no changes needed") systematically missed verbose decline-with-restatement responses: outputs that restate the original content while declining to change it. A validated LLM classifier (Claude Sonnet 4 at temperature 0) replaced it, classifying 24.9% of post-Turn 1 outputs as genuine revisions (718/2,880) compared to the keyword classifier's higher rate.

Ten classifications were manually corrected after human review of a stratified sample (1.4% correction rate). The classifier gap itself is a finding: models frequently produce verbose responses that mimic revision without substantive content change.

3.3 Blind Evaluation

Each model-turn output is scored by a blind evaluator on a six-level quality scale:

1. **Inadequate:** Does not address the task or is off-topic.
2. **Incomplete:** Addresses the task but is missing requested components.
3. **Functional:** All components present but with clear weaknesses.
4. **Sufficient:** All components present and competently executed.
5. **Polished:** Well-executed with thoughtfulness beyond the minimum.
6. **Overdone:** Adds unrequested complexity or has drifted from the original ask.

Level 4 ("Sufficient") is the threshold for task completion.

Scale Non-Monotonicity and 6→2 Recode. Level 6 ("Overdone") represents over-elaboration or drift from the task; it is *not* quality above Level 5. Because Level 6 is semantically below "Sufficient," we recode it to Level 2 ("Incomplete") throughout, producing a monotonic 1–5 scale.

Judge Calibration. To select the evaluator, we drew a stratified sample of 64 responses (balanced across model, domain, and turn) and had three human raters independently score each on the same six-level scale. We then ran each of six candidate models as evaluator on the same samples and selected the model with highest human-mean correlation. Claude Sonnet 4 was selected (Spearman $r = 0.505$, $p < 0.001$, QW $\kappa = 0.526$). The next-best candidate (DeepSeek V4) achieved $r = 0.340$; GPT-4o achieved only $r = 0.140$.

Human Validation. Three raters independently scored 64 stratified samples on the same scale (with 6→2 recode applied before analysis). Pairwise quadratic-weighted Cohen's κ ranges from 0.406 to 0.603; Krippendorff's $\alpha = 0.529$ (interval scale, 3 raters). Binary agreement on the sufficient/not-sufficient threshold (≥ 4) ranges from 76.6% to 82.8% across rater pairs; within-one-level agreement ranges from 81.2% to 95.3%. We note that the binary threshold, which drives the key analyses (revision-despite-sufficiency, DRP), is the better-validated metric; ordinal means should be interpreted with the moderate reliability in mind.

3.4 Meta-Commentary Stripping

Models frequently add boilerplate preambles ("Here's my revised version...") and postambles ("Let me know if you'd like any changes") to revision outputs. In our 50 balanced-panel pairs, 84% of revision-side outputs contain at least one such wrapper, compared to 14% of Turn 1 outputs. This asymmetry creates a measurement confound: LLM judges can identify the revision side by its meta-wrapping, and pointwise evaluators may trigger the "Overdone" criterion on boilerplate rather than content.

To control for this, we strip meta-commentary from all 3,600 outputs using validated regex patterns matching common preamble and postamble phrases. Of the 3,600 outputs, 1,200 (33.3%) were modified by stripping and rescored by Claude Sonnet 4 at temperature 0 using the identical evaluation prompt. Validation: on the 50 balanced-panel pairs, two human annotators rating stripped content agreed with the LLM judge at $\kappa = 0.569$, compared to near-chance agreement (κ between -0.07 and $+0.02$) on unstripped content. All quality results in Section 4 report stripped scores as primary estimates.

3.5 Edge Case Framework

To guard against aggregation artifacts, we apply four controls before interpreting quality trajectories:

1. **Balanced-panel subset:** The 50 trials (6.9%) with genuine revision (per the validated LLM classifier) at all turns T2–T5, enabling paired comparison without missing data. This subset conditions on a post-treatment variable (revision persistence), introducing potential selection bias; we triangulate it against pooled estimates below.
2. **Per-model trajectories:** Decomposing aggregate trends by model. The balanced panel

is 90% Llama (45/50); only Llama has sufficient power for within-model inference.

3. **T1 quality stratification:** Separating trials starting below the sufficiency threshold (12.4%) from those already sufficient (87.6%, after 6→2 recode).

4. **Compositional audit:** Checking whether shifting model proportions in the revision-only pool inflate apparent decline.

3.6 Metrics

- **Revision yield:** Mean evaluator level at each turn, restricted to revisions (meta-responses excluded).
- **Diminishing Return Point (DRP):** The first turn at which marginal quality gain is ≤ 0.
- **Revision-despite-sufficiency:** Fraction of turns where the evaluator rated the output ≥ 4 but the model revised at the next turn.
- **Overcorrection Score (OCS):** Composite of excess rounds beyond DRP, wasted token fraction, and quality regression magnitude.
- **Revision tax:** Extra tokens spent past the quality-optimal turn (t^*) as a percentage of optimal tokens: tax $= (T_{\text{full}} - T_{t^*})/T_{t^*} \times 100$.
- **Semantic similarity:** Cosine similarity between sentence embeddings (all-MiniLM-L6-v2) of the Turn 1 output and each subsequent turn.
- **Edit ratio:** Defined as $1 - \text{LCS}(r_t, r_{t-1})/\max(|r_t|, |r_{t-1}|)$, where LCS is the longest common subsequence measured in whitespace-delimited tokens.

3.7 Sub-experiments

Targeted Feedback. For outputs rated level 1–3, a second model instance receives the output plus a specific critique and produces a targeted revision, which is then blind-evaluated. This tests whether specific feedback outperforms the generic revision probe ($n = 177$ paired comparisons after filtering to cases where the generic next-turn output is a genuine revision, not a meta-response).

Self-Reflection. After the five-turn conversation, the same model is shown all its versions and asked which turn it would recommend the user use ($n = 720$).

Reversibility. Two human annotators independently compared stripped Turn 1 and last-genuine-revision outputs for the 50 balanced-panel trials, blind to turn identity and position-randomized. Inter-annotator agreement: $\kappa = 0.703$ ($n = 50$). Humans chose Turn 1 in 56.2% of non-tie decisions (41/73, 95% CI [45.2%, 67.1%]); the confidence interval includes 50%, so this directional preference is not statistically distinguishable from chance. We set a success bar in advance of $\geq$65% Turn 1 preference with a 95% CI excluding 50%; this bar was not cleared. This is itself informative: the pointwise quality degradation (-0.74 levels) is real but not large enough to reliably flip a blind pairwise preference once meta-commentary is removed. An LLM judge (Claude Sonnet 4) was also run on both stripped and unstripped versions for validation of the meta-commentary stripping procedure (see Section 3.4).

3.8 Preliminary Gate Experiments

Two preliminary experiments (reported fully in the appendix) establish that the revision decision is controlled by prompt phrasing, not quality assessment:

Study 1 (3,840 trials): A fully factorial design crossing 2 probe types $\times$ 8 threshold levels $\times$ 2 framings $\times$ 3 models $\times$ 8 scenarios $\times$ 5 runs. Under a revision-implying probe ("Can this be improved?"), models revise 99.9% of the time regardless of stated quality threshold. Under an evaluative probe, revision drops to 0.3–38% depending on model. Thresholds modulate revision intensity only after the gate fires (Spearman $\rho = -0.14$ to -0.50).

Study 2 (1,728 trials): A momentum experiment showing that 1–3 prior revision rounds shift the revision gate on a subsequent evaluative probe. GPT-4o jumps from 31% to 98% revision with a single prior round; Claude and Gemini resist. Reverse momentum (one affirming turn) suppresses revision to near-zero. These results motivate Study 3's balanced probe design: we use a probe that neither implies revision nor evaluation, making any observed degradation a conservative estimate of real-world behavior where revision-implying prompts dominate.

4 Results

We report meta-commentary-stripped quality scores as our primary estimates throughout;

Turn	Genuine revisions	Rate
T2	283/720	39.3%
T3	185/720	25.7%
T4	154/720	21.4%
T5	96/720	13.3%

Table 2: Genuine revision rate by turn. Classifications from a validated LLM classifier (Claude Sonnet 4 at temperature 0; see Section 3.2). The majority of post-Turn 1 outputs are meta-responses, not revisions.

unstripped values are given in parentheses for transparency. Section 3.4 describes the stripping procedure and its validation.

4.1 Models Mostly Decline to Revise

When prompted with undirected revision requests ("Can you improve this?"), models produce genuine revised content at a declining rate across turns (Table 2). By Turn 5, only 13.3% of trials contain a genuine revision; the remaining 86.7% are meta-responses: polite declines, restatements of the original, or commentary about the work rather than new task content.

Decline behavior is verbose and content-shaped. The distinction between genuine revisions and declines is non-trivial to automate. A keyword-based classifier (checking for phrases like "no changes needed") yielded 135 "balanced panel" trials with apparent revision at all five turns. A validated LLM classifier, checked against human judgment on a stratified sample, classified only 24.9% of post-Turn 1 outputs as genuine revisions (718/2,880), yielding 50 balanced-panel trials. The 85-trial gap reflects verbose decline-with-restatement responses: outputs that restate the original content (sometimes verbatim) while declining to change it, giving the surface appearance of revision without substantive modification.

Survival varies by model. Genuine-revision survival to Turn 5 ranges from 0.8% (Gemini 2.5 Flash: 1/120 trials) to 53.3% (Llama 3.3 70B: 64/120). Gemini declines in 98.3% of post-Turn 1 observations, the most extreme case. The balanced panel of 50 trials with genuine revision at all turns is composed of 45 Llama trials (90%), 3 Qwen, and 2 Claude; GPT-4o, DeepSeek, and Gemini contribute zero trials each.

Probe phrasing, not output quality, controls whether a model revises. Across five probe wordings tested in Study 1 (3 models; n = 3,932; see §3.8 and Appendix A), the revision gate is bimodal. The two probes that request change ("Can this be improved?" and "Is there anything you would change?") trigger near-universal revision (99.9%, n = 1,920; 100%, n = 18). The three probes that request assessment ("Take another look and let me know if it's ready," "Does it meet the bar?," and "What do you think?") produce single-digit to low-double-digit revision rates (23.2%, n = 1,920; 12.5%, n = 24; 2.0%, n = 50). No probe falls in the 40–99% range; the gate is semantic, not graded. Study 3's balanced probe ("Would you like to keep this as your final version, or would you like to revise it?"), measured with a different instrument (validated GENUINE/META classifier rather than the Study 1 evaluator gate), lands in the assessment range at 39.3% at Turn 2 (n = 720), consistent with this pattern.

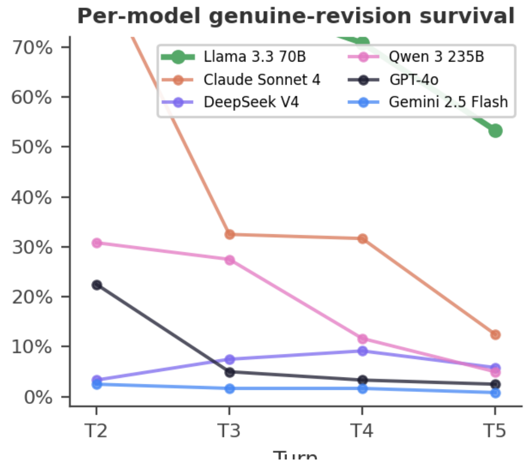

Figure 1: Per-model genuine-revision survival. Fraction of each model's 120 trials still producing genuine revised content at each turn. Llama 3.3 70B sustains revision through Turn 5 (53.3%); all other models drop below 15% by Turn 5. Gemini 2.5 Flash is near floor throughout (0.8% at T5).

4.2 Quality Degrades Under Undirected Revision

The revision cliff. In the 50 balanced-panel trials (genuine revision at all turns), mean quality on stripped content drops from 3.66 at Turn 1 to 2.92 at Turn 5, a decline of −0.74 levels (Wilcoxon signed-rank $p = 1.01 \times 10^{-4}$; unstripped: −0.94, $p = 3.3 \times 10^{-6}$; Table 3). This is a drop from between "Functional" and "Sufficient" to below "Incomplete."

	Stripped	(Unstripped)	Meta share
All balanced (n=50)	$\mathbf{-0.74}$ ($p = 1.01 \times 10^{-4}$)	(-0.94, $p = 3.13 \times 10^{-6}$)	21%
Llama only (n=45)	$\mathbf{-0.69}$ ($p = 3.76 \times 10^{-4}$)	(-0.82, $p = 1.28 \times 10^{-5}$)	16%

Table 3: Quality cliff (Turn 1 minus Turn 5) on balanced-panel trials. Stripped values are primary; unstripped values in parentheses. Meta share indicates the fraction of the unstripped decline attributable to meta-commentary artifact rather than content degradation.

Turn	Stripped	(Unstripped)	Shift	n
T1	4.11	(4.11)	0	720
T2	3.66	(3.25)	+0.41	283
T3	3.40	(3.01)	+0.39	185
T4	3.34	(2.97)	+0.37	154
T5	3.07	(2.85)	+0.22	96
Δ (T1–T5)	$\mathbf{-1.04}$	(-1.26)		

Table 4: Pooled mean quality by turn, restricted to genuine revisions (GENUINE-only at T2–T5, all 720 at T1). Stripped scores are primary. n declines as models exit the revision pool.

Per-model caveats. The balanced panel is dominated by Llama 3.3 70B ($n = 45$), which is the only model with sufficient power to detect a within-model cliff. Its stripped cliff of -0.69 is significant at $p = 3.76 \times 10^{-4}$ ($r = 0.53$). Qwen ($n = 3$, $\Delta = -3.00$) and Claude ($n = 2$, $\Delta = -0.50$) show directional degradation but are too small for inference. GPT-4o, DeepSeek, and Gemini contribute zero balanced-panel trials, not because they maintain quality, but because they stop producing genuine revisions before Turn 5. We do not claim "all models degrade"; we claim that among models that continue revising, the only powered comparison (Llama, $n = 45$) shows significant degradation.

Over-elaboration rises with revision. Among genuine revisions, the Level 6 ("Overdone") rate rises from 5.7% at T1 to 47.9% at T5 (34.4% after meta-commentary stripping), confirming that the evaluator assigns "Overdone" increasingly at later turns. This recode corrects an artifact in which Llama 3.3 70B appeared to improve with revision: the apparent improvement was driven by Level 6 scores that inflated the raw mean at later turns.

Pooled trajectory across all genuine revisions. Relaxing the balanced-panel requirement and pooling all genuine revisions (with shifting n at each turn), mean quality on stripped content declines from 4.11 at Turn 1 to 3.07 at Turn 5 ($\Delta = -1.04$; unstripped: -1.26; Table 4). This trajectory has known compositional bias: models with higher decline rates (Gemini, GPT-4o) exit the revision pool earlier, leaving models with lower decline rates (Llama) overrepresented at later turns. The direction is nevertheless consistent with the balanced-panel estimate.

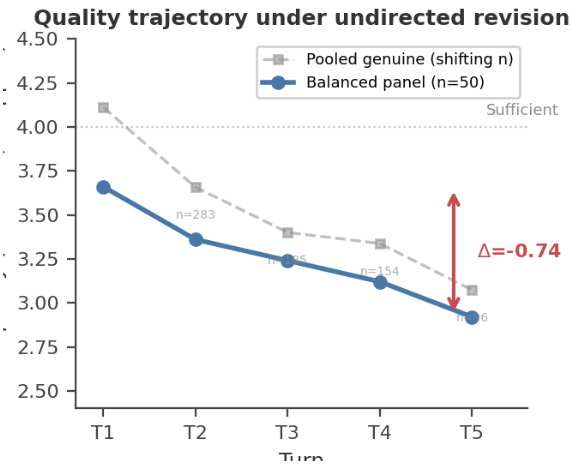

Figure 2: Quality trajectory under undirected revision (stripped scores). Solid blue: balanced panel ($n = 50$, genuine revision at all turns), showing the headline cliff of $\Delta = -0.74$ (3.66→2.92). Dashed gray: pooled genuine-only trajectory (shifting n, compositional bias from differential dropout), $\Delta = -1.04$. Both estimators decline monotonically. The dashed horizontal line marks the sufficiency threshold (level 4).

Domain variation. With probe phrasing held constant in Study 3, model identity explains far more variation in genuine-revision rate than task domain: the model range is 71.9 percentage points (Llama 3.3 70B 73.5% to Gemini 2.5 Flash 1.7%, $n = 480$ per model) while the domain range is 13.5 percentage points ($n = 576$ per domain). Which model is asked to revise matters roughly 4–5$\times$ more than what it is asked to revise. Within that smaller domain effect, code tasks elicit significantly more genuine revision: 34.0% of post-Turn 1 code outputs are genuine revisions vs. 20.5–25.2% for the four other domains ($\chi^2 = 35.63$, $df = 4$, $p = 3.4 \times 10^{-7}$). Code is the highest-revision domain for every model except DeepSeek ($n = 96$ per domain-model cell), which is near floor on all domains (3–10%). This is consis-

Domain	Stripped Δ	(Unstripped Δ)	Shift	n(T5)
analysis	−1.16	(−1.21)	+0.05	17
code	−1.05	(−1.15)	+0.10	30
creative	−0.82	(−1.22)	+0.40	19
data_logic	−1.01	(−1.35)	+0.34	17
writing	−1.06	(−1.31)	+0.25	13

Table 5: Quality decline (T1–T5) by domain on stripped content. All domains show negative deltas; 4/5 are significant ($p < 0.05$), data_logic marginal ($p = 0.058$). Per-domain n(T5) limits precision.

Condition	Targeted mean	Generic mean	Δ	p
Both stripped	4.68	3.53	**+1.16**	$5.7 \times 10^{-}$
(Both unstripped)	(4.68)	(4.43)	(+0.25)	$(3.8 \times 10^{-}$

Table 6: Targeted vs. generic revision quality ($n = 177$ matched pairs). Stripped scores are primary. The unstripped comparison understates the effect because meta-commentary inflates generic revision scores.

Model	n	Stripped Δ	p
Llama 3.3 70B	71	+1.62	9.2×10^{-12}
Qwen 3 235B	21	+1.48	4.5×10^{-4}
GPT-4o	12	+1.33	7.8×10^{-3}
DeepSeek V4	19	+1.05	1.2×10^{-2}
Claude Sonnet 4	51	+0.31	1.2×10^{-2}
Gemini 2.5 Flash	3	+2.33	– ($n < 5$)

Table 7: Targeted-feedback advantage by model on stripped content. All five powered models show significant improvement.

tent with the targeted-feedback mechanism (Section 4.3): code tasks carry built-in directed signals (a failing test, a type error, a clear specification to check against) that function as implicit critique, the same evaluative input that targeted feedback provides explicitly. The domain effect is additive rather than interactive: code is boosted roughly 10–15 percentage points above each model's own baseline revision rate, and domain×model cells ($n = 96$) are large enough to detect a crossover interaction if one existed; none does.

All five domains show negative stripped deltas on the pooled genuine trajectory (Table 5), and four of five reach significance on paired Wilcoxon tests ($p < 0.05$; data_logic is marginal at $p = 0.058$). Creative writing shows the largest sensitivity to stripping (stripped $\Delta = -0.82$ vs. unstripped -1.22), consistent with higher meta-commentary prevalence in creative tasks. Code shows the least sensitivity (-1.05 vs. -1.15). However, per-domain Turn 5 cells range from $n = 13$ (writing) to $n = 30$ (code), and balanced-panel cells from $n = 5$ (writing) to $n = 22$ (code); only code clears $n \geq 10$ in the balanced panel. The aggregate degradation pattern is well-powered; per-domain cliff magnitudes should be read as directional estimates, not precise per-domain measurements. (Creative writing shows the smallest balanced-panel cliff estimate, $\Delta = -0.17$ at $n = 6$, but this cell is too small to interpret; whether creative tasks genuinely resist degradation under undirected revision is an open question requiring more genuine-revision data.)

Revision despite sufficiency. Among all (trial, turn) pairs where the evaluator rated the output as sufficient (level ≥ 4), the model produced a genuine revision at the next turn 39.2% of the time (368/938 sufficient turns, 95% CI [36.1%, 42.3%]; stripped: 39.6%, 411/1,038). Nearly two in five sufficient outputs are followed by an unnecessary revision.

4.3 Targeted Feedback Restores Quality

The preceding sections show that undirected revision degrades quality. Directed revision, providing specific critique before requesting a revision, substantially improves it. On stripped content, targeted revisions score 1.16 levels above the model's own generic next-turn revision ($p = 5.7 \times 10^{-19}$, $n = 177$; unstripped: $+0.25$, $p = 0.004$; Table 6). The unstripped comparison understates the real effect because meta-commentary on generic revisions inflates their scores; targeted revisions carry minimal meta-commentary (9/177 = 5%).

This effect is significant for all five models with sufficient sample size (Table 7). On unstripped text, only Claude and DeepSeek showed significant effects; stripping reveals that Llama, Qwen, and GPT-4o also benefit substantially, but their generic baselines were inflated by meta-commentary.

The contrast between undirected and directed revision is the central practical finding: undirected revision degrades quality by 0.74 levels over five turns, while a single directed revision improves it by 1.16 levels. The problem is not revision capacity but the absence of evaluative direction.

4.4 The Revision Tax

The quality-optimal stopping point (t^*) is Turn 1 for all six models: no model benefits from undirected revision on the genuine-revision quality tra-

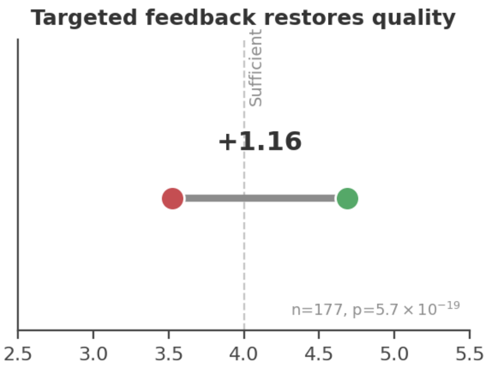

Figure 3: Targeted feedback restores quality. Generic undirected revision produces mean quality 3.53 (stripped); targeted feedback with specific critique produces 4.68, a gain of +1.16 levels ($n = 177$, $p = 5.7 \times 10^{-19}$). The dashed line marks the sufficiency threshold.

Model	t^*	Tax %	Waste %
Gemini 2.5 Flash	T1	30.6	23.4
DeepSeek V4	T1	118.9	54.3
GPT-4o	T1	125.8	55.7
Qwen 3 235B	T1	198.6	66.5
Claude Sonnet 4	T1	251.6	71.6
Llama 3.3 70B	T1	436.0	81.3
Aggregate	T1	164.2	62.1

Table 8: Revision tax by model. $t^* =$ Turn 1 for all models. Tax % = (waste tokens/baseline tokens) × 100. Waste % = waste tokens/total tokens. Meta-response tokens are counted as waste (Interpretation A).

jectory. All tokens generated after Turn 1 are therefore waste under a quality-maximizing criterion. Our revision tax is adjacent to the independently coined "YapTax" of Borisov et al. (2026), which measures single-turn over-generation on brevity-ideal prompts (excess tokens above a minimal-sufficient baseline × output price); our metric instead captures multi-turn revision waste, where the model is asked to improve an already-sufficient output and the excess accumulates across turns.

Across all 720 trials, 62.1% of output tokens are spent past t^* (aggregate tax: 164.2%; Table 8). Per-model waste ranges from 23.4% (Gemini, which declines early and generates few post-T1 tokens) to 81.3% (Llama, which continues revising and generates substantial post-T1 content). Translated to deployment cost at 2025 API prices, these post-t^* tokens project to between \$323 per year (Gemini 2.5 Flash) and \$65,678 per year (Claude Sonnet 4) for a 500-person organization; the range is dominated by model choice, combining a roughly 37-fold spread in output price with the models' differing revision volumes, and should be read as an order-of-magnitude indication rather than a precise forecast.

5 Discussion

Evaluate, then direct. The central practical finding is that undirected revision and directed revision are not two points on a spectrum but opposites. Asked to improve its work without guidance, a model degrades it; given a single specific critique, the same model improves it substantially. The failure is therefore not a limit on what models can do, but on what they will do absent direction. This has a likely origin in how models are trained: reinforcement from human feedback rewards visible effort, and a model that revises when asked is rated more favorably than one that correctly declines, regardless of whether the revision helped. Over enough such examples, the model learns that the safe response to a revision request is to revise. The remedy follows directly. For users, the rule is to withhold the request to revise until you can name what is wrong; "make it better" is not an instruction a model can act on well, because it carries no information about what better means. For systems that automate revision, the implication is architectural: place an evaluation step before the revision step, so that revision is triggered by a specific identified weakness rather than by default. A model's revision capacity is worth invoking only once there is something to direct it at.

The cost of undirected revision is real and mostly hidden. Because the failure is quiet, it is easy to underpay attention to. A user who cannot tell a degraded draft from an improved one absorbs the loss without noticing, and the tokens spent reaching that worse draft are billed all the same. Two costs compound here. The first is the wasted generation: across our trials the quality-optimal stopping point is the first turn for every model, so every token spent on undirected revision is, on average, spent making the output worse or no better, 62% of output tokens in our experiment fall past that point. The second is subtler and larger in aggregate: most models do not revise at all but answer with verbose meta-responses, paragraphs of acknowledgment and restatement that change nothing yet still consume tokens and still leave the user with a draft they cannot evaluate. Both costs scale with exactly the trend driving model adoption. Enterprise inference spending is climbing steeply, and the multi-turn, agentic workflows responsible for that climb are where undirected revision loops live, an autonomous system that revises by default, with no human to notice the draft is getting worse, compounds the waste at machine speed. Iteration caps and turn limits, the usual engineering guardrails, bound the spending but not the underlying error: they stop the loop without ever asking whether the revision was worth running. The failure is already visible in deployed agentic systems: public issue trackers document agents stuck in infinite loops editing the same file (VSCode Issue #257885, Claude Code Issue #27281) (Anthropic, 2025b).

Revision robustness deserves to be measured. Current evaluation looks in the wrong place. Benchmarks score a model's first answer, its single-turn quality on a fixed task, and models have become very good at this; in our data the first draft is already sufficient the large majority of the time. But the property that makes a model good on a single-turn benchmark, strong first-draft quality, is not the property that makes it safe across a multi-turn revision: the willingness to stop, to recognize that an output needs no further change. Nothing in standard evaluation measures that willingness, and our results show the two properties come apart, models that produce excellent first drafts will still degrade them when asked to revise. We propose that revision robustness, whether a model can be handed its own sufficient output and decline to make it worse, be evaluated alongside first-turn capability. Measuring it, however, requires care, because models distort the measurement. The meta-commentary they wrap around revisions, the prefaces and postscripts addressed to the user rather than the task, misleads an automated judge in two opposing directions at once: it inflates the judge's preference for the earlier draft while depressing the measured quality of the later one. An evaluation that does not strip this commentary will reach confident and contradictory conclusions about the same outputs. This is not specific to our setup. Any study that compares first-draft and revised model outputs with an LLM judge inherits the same confound, and controlling for it should be standard practice.

6 Limitations

Several limitations qualify these findings. First, the balanced panel is dominated by Llama 3.3 70B ($n = 45$ of 50); other models exit the revision pool too quickly for powered within-model comparison. Second, the evaluator (Claude Sonnet 4) judges outputs from all models including itself; we mitigate this through six-model calibration selecting the highest human-correlation judge (Spearman $r = 0.505$, QW $\kappa = 0.526$). Third, inter-rater reliability is moderate (quadratic-weighted κ = 0.41–0.60 across three rater pairs; Krippendorff's $\alpha = 0.529$, $n = 64$). Fourth, all trials used temperature 1.0; revision behavior at lower temperatures may differ. Fifth, our 40 tasks represent a convenience sample across five domains rather than a stratified probability sample. Sixth, although the balanced probe was designed to be neutral, it may still subtly prime revision by raising the possibility; a pure observation condition (no probe) would provide a stronger baseline. Seventh, our per-domain quality estimates rest on small per-domain samples: Turn-5 genuine-revision cells range from 13 to 30 trials, and only the code domain clears ten trials in the balanced panel, so per-domain degradation magnitudes should be read as directional rather than precise. Eighth, our reversibility result is near chance: after stripping meta-commentary, blind human preference for the first draft is 56%, with a confidence interval that includes 50%, and a pre-set success bar of 65% that was not cleared, so we do not claim humans can reliably distinguish first from revised drafts.

7 Conclusion

When LLMs are given repeated undirected revision opportunities, most decline to genuinely revise (75% of post-Turn 1 outputs are meta-responses), and those that do revise get worse. The revision cliff is real (-0.74 levels on stripped content, balanced panel, $p = 1.01 \times 10^{-4}$), consistent across task domains, and costly (62.1% of output tokens wasted, with the quality-optimal stopping point at Turn 1 for all six models tested).

The fix is targeted feedback, not more turns. When told what to fix, models produce revisions 1.16 quality levels higher than generic iteration ($p = 5.7 \times 10^{-19}$), an effect that holds across all five powered models. The most cost-effective intervention is not suppressing revision but directing it: evaluate first, then provide specific critique only when revision is warranted. For organizations, the implication is to implement quality gates before revision loops. For users, the implication is to stop asking "can this be improved?" and start specifying what is wrong. For the field, the implication is that multi-turn revision robustness deserves evaluation alongside single-turn capability.

Ethics Statement

This study includes human evaluation by three raters (the first author and two research assistants) who scored AI-generated outputs on a quality scale. This evaluation involved reading and rating synthetic text only; no personal data was collected, and raters participated voluntarily. All data is generated by commercial LLM APIs in response to synthetic scenarios spanning code, data logic, analysis, and writing tasks. The scenarios contain no sensitive content. We do not release model outputs that could be mistaken for human-written text. Pricing data is sourced from publicly available API documentation and subscription pages as of June 2026.

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A Study 1: The Revision Gate

Study 1 tests whether user-stated quality thresholds constrain the decision to revise. In a fully factorial experiment (8 scenarios $\times$ 16 threshold conditions $\times$ 2 probes $\times$ 3 models $\times$ 5 runs $=$ 3,840 trials), the follow-up probe's phrasing dominates the revision gate. Under the leading probe ("Can this be improved?"), models revised 99.9% of the time regardless of threshold. Under the evaluative probe ("Take another look and let me know if it's ready"), revision rates varied by model: Gemini 2.5 Flash declined 99.7%, GPT-4o 68.6%, Claude Sonnet 4 62.0%.

Chi-squared tests confirm the revision gate distribution differs significantly by probe type ($\chi^2 > 788$, $p < 0.0001$ for all models). Quality thresholds do not significantly affect the gate (Kruskal-Wallis $p > 0.40$ for all models).

Within the leading-probe condition, higher thresholds produce moderately less overcorrection (Spearman ρ: Gemini -0.50, Claude -0.35, GPT-4o -0.14), but this calibration is locked behind the phrasing-sensitive gate.

Predictor	Model A	Model B	Model C
is_leading	3.47^{***}	3.94^{***}	—
threshold	-0.93^{***}	-1.04^{***}	-1.65^{***}
is_qualitative	-0.74^{***}	-0.85^{***}	-1.05^{***}
Claude Sonnet	0.10	0.15	0.33^{**}
Gemini Flash	0.42^{***}	0.52^{***}	1.15^{***}
Scenario FEs	No	Yes	Yes
N	3,840	3,840	1,920
AIC	5,319	4,839	3,545

Table 9: Ordinal regression (overcorrection $\sim$ predictors). Model C: leading-probe only. $^{**}p < 0.01$, $^{***}p < 0.001$. Reference: GPT-4o.

B Study 2: Momentum

Study 2 tests whether prior revision rounds shift the revision gate on a subsequent evaluative probe (1,728 trials). Without prior revision (dose 0), models revised 23.2%; after 1–3 prior rounds, this rises to 44.6% ($\chi^2 = 376.54$, $p < 0.0001$).

Model-level divergence is dramatic:

- GPT-4o: 31.4% $\rightarrow$ 98.4% at dose 1 (near-total sycophantic shift)

- Claude Sonnet 4: 38.0% $\to$ 24.5% at dose 3 (declines under momentum)
- Gemini 2.5 Flash: 0.3% $\to$ 12.8% at dose 1 (modest shift from near-zero)

The momentum effect is a step function: the entire shift occurs between dose 0 and dose 1, with no additional gain at higher doses. Threshold level does not interact with momentum ($p = 0.51$).

Reverse Momentum. A single turn of affirming feedback ("This looks great, no changes needed") suppresses full revision to near-zero across all models: Gemini 0.0%, GPT-4o 1.0%, Claude 21.9% (all minor suggestions). The gate tracks the most recent conversational signal in both directions.

C Pipeline Diagrams

D Supplementary Figures

E Full Statistical Tables

E.1 Pairwise Threshold Comparisons (Study 1)

Model	**Comparison**	U	p_{adj}	r
Gemini (num.)	70 vs. 100	4935	<0.001	−0.42
Gemini (num.)	85 vs. 100	4238	<0.001	−0.32
Gemini (num.)	0 vs. 100	4755	0.002	−0.25
Gemini (qual.)	0 vs. 100	4084	0.002	−0.28
Claude (num.)	85 vs. 100	4146	<0.001	−0.30
Claude (qual.)	85 vs. 100	3880	0.024	−0.21

Table 10: Significant pairwise comparisons (Bonferroni-corrected, $q < 0.05$).

E.2 Power Analysis

Post-hoc power analysis at $\alpha = 0.05$, 80% power:

- **Study 1 RQ1** (N = 1,920 per probe): MDE for proportion difference = 0.032. Observed ≈ 0.75. Massively overpowered.
- **Study 1 RQ2** (N = 1,920 leading-probe): MDE for $|\rho| = 0.064$. All observed correlations exceed this.
- **Study 3** (N = 720 trials, balanced panel $n = 50$): Panel-level MDE = 0.5 levels. Observed $\Delta = 0.94$ unstripped (0.74 after meta-commentary stripping), both exceeding the MDE. The effective sample for the paired test is the balanced panel ($n = 50$ trials with genuine revision at all turns), which yields $p = 3.32 \times 10^{-6}$ (Wilcoxon, $r = 0.658$) on the unstripped cliff; the stripped cliff ($\Delta = 0.74$) has $p = 1.01 \times 10^{-4}$ ($r = 0.55$).

F Judge Calibration Details

Candidate Judge	**Spearman** r	**QW** κ
Claude Sonnet 4	0.505	0.526
DeepSeek V4	0.340	0.398
Qwen 3 235B	0.272	0.017
Llama 3.3 70B	0.181	0.022
GPT-4o	0.140	0.065
Gemini 2.5 Flash	−0.036	−0.026

Table 11: Judge calibration results ($n = 64$ stratified samples each; Spearman correlation against the human rater mean). Claude Sonnet 4 was selected as the Study 3 evaluator, with the highest human correlation of the six candidates ($r = 0.505$, $p = 2.1 \times 10^{-5}$). Gemini 2.5 Flash was anti-correlated with human judgment.

F.1 Human Rater Agreement

Rater Pair	**QW** κ	**Binary** (≥ 4)	**Within-1**
Liam – Sophie	0.578	78.1% (50/64)	95.3% (61/64)
Sophie – Troy	0.603	82.8% (53/64)	87.5% (56/64)
Liam – Troy	0.406	76.6% (49/64)	81.2% (52/64)
All 3 (Krippendorff's α)	0.529	—	—
Evaluator – Human Mean	0.526	—	—

Table 12: Human rater agreement on 64 stratified samples (6→2 recode applied before analysis, 3 raters). QW κ: quadratic-weighted Cohen's κ on the 1–5 scale. Binary: both raters agree on sufficient (≥ 4) vs. not. Within-1: ratings differ by ≤ 1 level. Evaluator–Human Mean: Spearman $r = 0.505$ ($p < 0.001$).

G Study 1 Inter-Rater Reliability

Dimension	**QW** κ	**Interpretation**
Revision magnitude	0.844	Excellent
Revision value	0.690	Good
Overcorrection	0.609	Acceptable
Threshold alignment	0.556	Marginal

Table 13: Inter-rater reliability for Study 1 (GPT-4o vs. Claude Sonnet 4, $n = 60$).

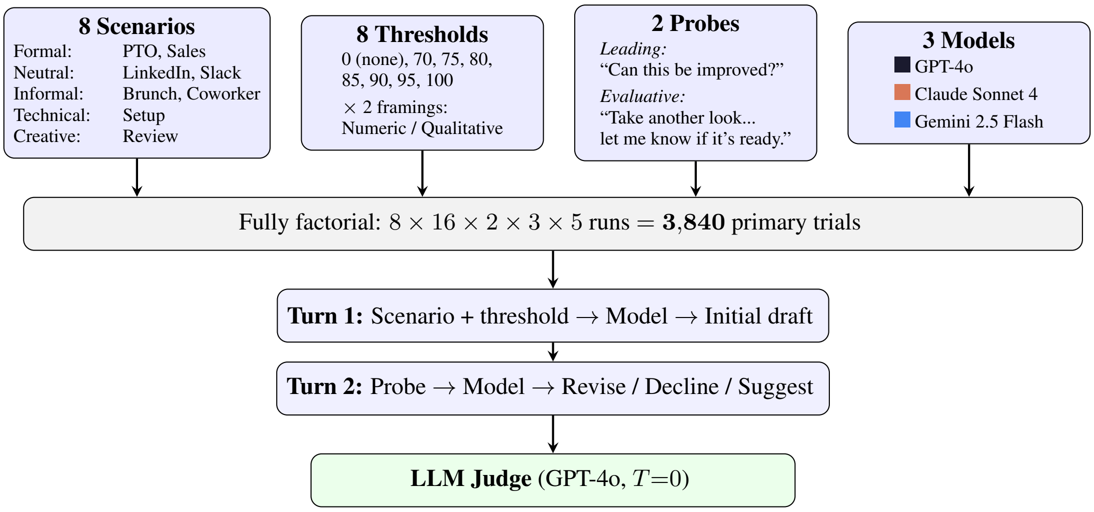

Figure 4: Study 1 experimental pipeline. Four factors crossed in a fully factorial design produce 3,840 trials.

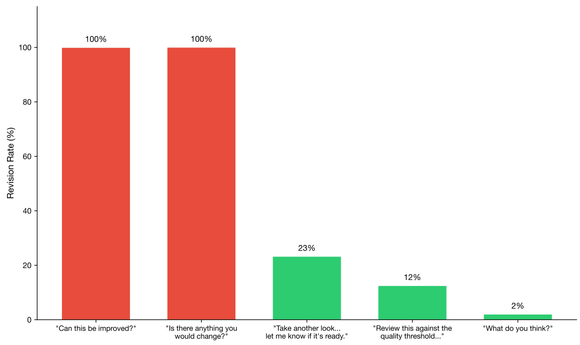

Figure 5: Compliance cliff (Study 1 pilot). Revision rates drop from 100% to $\leq$25% across five pilot probes with no intermediate values.

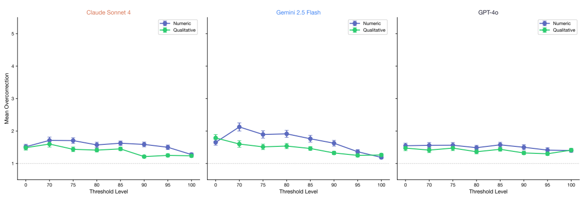

Figure 6: Overcorrection by threshold level within leading-probe trials (Study 1).

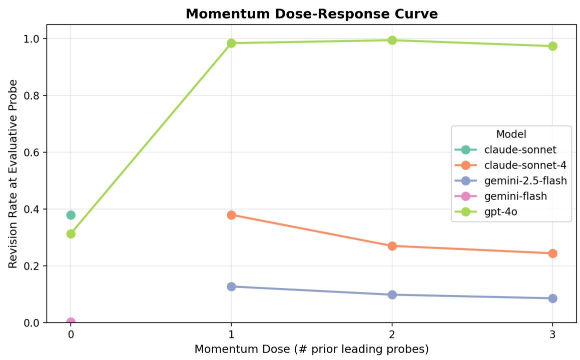

Figure 7: Revision rate by momentum dose (Study 2). The momentum effect is a step function driven by GPT-4o.

Dimension	**QW κ**	**% Agree**	**% Within 1**
Revision magnitude	0.575	68.3	85.0
Revision value	0.669	76.7	88.3
Overcorrection	0.392	71.7	90.0
Threshold alignment	0.339	38.3	85.0

Table 14: Inter-rater reliability for Study 2 ($n = 60$).